Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

B. E. : EST
Course Name: Computer Architecture and Organization
Duration: 3 Hours, M. Marks: 44

Course Code: UCS510 / UCS319 / ULC303
Date & Time: 23/07/2026, 2:00 PM
Faculty: Dr. Abdul Kadir

Note: All questions carry equal marks (11 marks each). **Attempt any FOUR questions out of five.** Attempt all parts of a question at one place and in order. Assume suitable data wherever necessary and state your assumptions clearly.

Q.No.	Question	Marks	CO	BL
Q01.A	Simplify each of the following Boolean expressions to its minimum number of literals using **Boolean algebra laws and theorems**. Clearly state the law/theorem applied at every step (De Morgan's, distributive, absorption, consensus, complement, etc.). (a) $F_1 = AB + A'C + BC$ (b) $F_2 = (A + B)(A + B')(A' + C)$ (c) $F_3 = [(A + B)(A' + C)]' + A'C'$	3	CO1	BL3
Q01.B	Two 8-bit registers of a processor contain $R1$ = 1101 0110 and $R2$ = 0110 1011. Determine the content of $R1$ after each of the following **independent** microoperations, each applied on the **original** value of $R1$ (using $R2$ as the operand register wherever required). In each case, also name the logic/shift microoperation actually used: (a) Selective-set with $R2$ (b) Selective-complement with $R2$ (c) Mask (clear) operation with $R2$ (d) Circular shift left, followed by circular shift right	4	CO1	BL3
Q01.C	Design a **4-bit arithmetic circuit** that implements eight arithmetic microoperations (add, add with carry, subtract with borrow, subtract, transfer, increment, decrement and transfer with carry) using **binary adders and** 4×1 **multiplexers**. Draw the circuit for one typical stage and give the complete **function table** showing the values of the selection inputs S_1, S_0, the input carry C_{in}, the value of Y applied to the adder, and the resulting microoperation.	4	CO1	BL2

<table>
<tr><th>Q.No.</th><th>Question</th><th>Marks</th><th>CO</th><th>BL</th></tr>
<tr><td>Q02.A</td><td>Consider a 16-bit processor in which a two-word instruction is stored in the main memory starting at location 420, as shown in the figure. The first word of the instruction contains the mode field, which specifies the addressing mode, and the second word contains the address field. The instruction loads a value into the accumulator (AC). A source register $R1$ holds the value 800, and a base register BR holds the value 250. The memory is so initialized that location 1100 contains the value 1700, location 1099 contains 1699, location 1098 contains 1698, and so on (i.e. every location X contains the value $X + 600$).

<table>
<tr><th>Address</th><th>Memory</th></tr>
<tr><td>420</td><td>Mode</td></tr>
<tr><td>421</td><td>500 (Load AC)</td></tr>
<tr><td>422</td><td>⋮</td></tr>
</table>

Determine the effective address and the operand loaded into AC for each of the following addressing modes. Show the complete calculation in each case. (All numbers are in decimal.)
(a) Immediate
(b) Register indirect (using $R1$)
(c) Auto-decrement (using $R1$)
(d) Indirect relative
(e) Base register (using BR)</td><td>4</td><td>CO2</td><td>BL4</td></tr>
<tr><td>Q02.B</td><td>Answer the following with reference to cache memory:
(a) What is meant by the principle of locality of reference? Distinguish between temporal and spatial locality with one example each, and explain how a cache memory exploits both to reduce the average memory access time.
(b) Differentiate between the write-through and write-back cache write policies, stating one advantage and one disadvantage of each. Also explain what is meant by a dirty bit and in which of the two policies it is required.</td><td>3</td><td>CO4</td><td>BL2</td></tr>
<tr><td>Q02.C</td><td>Explain the interrupt cycle of the basic computer with the help of a neat flowchart. Clearly bring out the role of the flip-flops IEN, FGI, FGO and R, write the register-transfer (micro-operation) statements executed during the interrupt cycle, and explain how control returns to the interrupted program.</td><td>4</td><td>CO1</td><td>BL2</td></tr>
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Q.No.	Question	Marks	CO	BL
Q03.A	The 8-bit ALU of a processor operates on the operands A = 1001 0101 and B = 0110 1100, which are represented in **signed 2's complement** form. (a) Perform the operation $A - B$. Show the complete binary working and give the 8-bit result. (b) Determine the values of the four status bits C (carry), S (sign), Z (zero) and V (overflow) for the operation in part (b), justifying the value of V from the carries into and out of the sign-bit position. (c) State, with reason, whether each of the conditional branch instructions **BC** (branch if carry set), **BZ** (branch if zero), **BNC** (branch if carry clear) and **BV** (branch if overflow) would be taken after the operation in part (b). (d) Now perform the operation $A + B$ on the same operands and determine the four status bits again.	6	CO2	BL4
Q03.B	Consider the following sequence of instructions executed on a standard **5-stage instruction pipeline** (IF, ID, EX, MEM, WB). Assume that operands are read in the ID stage and results are written in the WB stage, and that the register file does **not** support split-cycle (write-then-read) access. I_1: LOAD R1, 0(R2) $(R1 \leftarrow M[R2])$ I_2: ADD R3, R1, R4 $(R3 \leftarrow R1 + R4)$ I_3: SUB R5, R3, R1 $(R5 \leftarrow R3 - R1)$ I_4: AND R6, R5, R3 $(R6 \leftarrow R5 \wedge R3)$ I_5: STORE R6, 4(R2) $(M[R2 + 4] \leftarrow R6)$ (a) Identify all the **RAW (true) data dependencies** present in the given sequence. (b) Draw the **space-time (Gantt) diagram** assuming **no operand forwarding**, clearly marking the stall cycles, and determine the total number of clock cycles required. (c) Draw the space-time diagram assuming **full operand forwarding** is implemented. Clearly indicate which forwarding path (EX/MEM $\rightarrow$ EX or MEM/WB $\rightarrow$ EX) supplies the operand for each dependent instruction, determine the total number of clock cycles, and identify the one dependency that still causes a stall even with forwarding, explaining why forwarding alone cannot eliminate it.	5	CO4	BL4

Q.No.	Question	Marks	CO	BL
Q04.A	A computer has a **32-bit byte-addressable** main memory and a cache of size **512 KB** with a block (line) size of **64 bytes**. Assume that each cache block carries one valid bit in addition to its tag. (a) Determine the total number of blocks (lines) in the cache and the number of bits in the **BLOCK OFFSET** field. (b) For a **direct-mapped** organization, determine the number of bits in the TAG and INDEX fields, the number of tag comparators required, and the total size of the tag directory in bits. (c) For an **8-way set-associative** organization, determine the number of sets, the number of bits in the TAG and SET fields, and the number of tag comparators required. (d) For a **fully associative** organization, determine the number of bits in the TAG field and the number of tag comparators required.	6	CO4	BL4
Q04.B	(a) What is an **associative (content-addressable) memory**? Draw its block diagram, explain the function of the argument register, key register and match register, and state one typical application in a memory system. (b) Explain **memory interleaving**. Distinguish between **low-order** and **high-order** interleaving with the help of diagrams, and state which of the two is better suited for improving the effective memory bandwidth and why.	5	CO3	BL2
Q05.A	What is **Direct Memory Access (DMA)**? Draw a neat block diagram of a DMA controller showing the address register, word count register, control register, data register and the DMA request (BR) / DMA acknowledge (BG) lines. Explain how a complete DMA transfer is initiated by the CPU, carried out by the controller and terminated, clearly bringing out how the CPU releases the address, data and control buses to the DMA controller. Also differentiate between the **burst (block) transfer mode** and the **cycle-stealing mode** of DMA operation.	5	CO4	BL2
Q05.B	Answer the following: (a) Explain the three modes of **I/O data transfer – programmed I/O**, **interrupt-initiated I/O** and **DMA**. For each mode, state how far the CPU is involved in the transfer and give one situation in which that mode is preferred. (b) A DMA controller transfers **16-bit words** to memory using **cycle stealing**. The words are assembled from a device that transmits characters (8 bits each) at a rate of **9600 characters per second**. The CPU fetches and executes instructions at an average rate of **2 million instructions per second**, and each instruction requires one memory access on the average. Determine the percentage by which the CPU is slowed down because of the DMA transfer.	6	CO4	BL3

End of Question Paper